

DEPARTMENT OF AIML

S.E. Capstone Project-I

**CareerMentor AI: An Autonomous Multilingual AI-
Powered Career Guidance & Management Agent**

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Introduction

In the contemporary technical education landscape, over 1.5 million engineering students graduate annually across Indian universities, yet independent national employability audits indicate that fewer than twenty percent possess the practical problem-solving and systems development competencies demanded by Tier-1 product organizations. Engineering undergraduates, particularly across tier-2 and tier-3 institutions, navigate critical career milestones in an environment plagued by acute institutional deficits, where student-to-counselor ratios routinely exceed 500:1. Simultaneously, modern corporate hiring pipelines have instituted algorithmic gatekeeping through Applicant Tracking Systems (ATS) that automatically discard over seventy-five percent of candidate submissions before human review due to unquantified accomplishments, structural parsing errors, and keyword omissions. Compounding this challenge, conventional career guidance tools remain strictly monolingual in English, imposing substantial cognitive friction on regional students who conceptualize engineering principles in Hindi or Marathi. CareerMentor AI is engineered as an autonomous, full-stack, multilingual artificial intelligence web agent that democratizes elite career coaching. Operating as a continuous 24/7 personal mentor, the platform seamlessly unifies multi-session conversational guidance, speech-to-text voice interaction, document-level resume abstract syntax analysis, collaborative human-plus-AI study planning, and proactive autonomous background check-ins across English, Hindi, Marathi, and Sanskrit, transforming abstract placement ambitions into measurable daily engineering velocity.

1.5M+ Graduates/Year

>80% lack verified software development competencies.

75% ATS Filter Rate

Resumes rejected due to formatting flaws & unquantified metrics.

4 Languages Supported

Localized mentorship across English, Hindi, Marathi, & Sanskrit.

Motivation

- **High Attrition in Static Online Learning:** Conventional internet roadmaps impose monolithic, rigid 8-hour daily study schedules that fail to accommodate real-world collegiate obligations, resulting in an 85%+ abandonment rate among self-directed students.
- **Absence of Behavioral & Technical Interview Feedback:** Freshers lack a risk-free, interactive evaluation environment to simulate high-stakes interviews, refine behavioral narratives via the STAR framework, or defend architectural trade-offs in distributed systems.
- **Severe Career Tooling Fragmentation:** Job seekers are forced to juggle five or more disconnected single-purpose utilities—standalone ATS text checkers, unstructured video playlists, calendar schedulers, and stateless chatbots—causing cognitive fatigue and loss of historical context.
- **Linguistic and Regional Exclusion:** The overwhelming majority of technical preparation resources cater exclusively to fluent English speakers, neglecting millions of talented regional engineering undergraduates who think in Hindi or Marathi.
- **Lack of Autonomous Accountability:** Passive educational platforms remain inert until the user manually logs in; engineering candidates require an active autonomous agent that proactively monitors remaining tasks and dispatches 2-hour interval check-ins.
- **Need for Science-Backed Focus Blocks:** Rigid 60-minute or 2-hour time barriers induce mental fatigue; implementing customizable focus budgets of 30, 45, 57 (optimal focus block), and 90 minutes drastically increases curriculum completion rates.

Problem Statement

Engineering undergraduates and freshers face severe placement barriers and career disorientation due to ATS-incompatible resumes, unstructured preparation paths, and the systemic absence of continuous, accessible, and multilingual technical mentorship. Existing solutions are either fragmented across disparate single-purpose utilities, cost-prohibitive, static in curriculum delivery, or strictly monolingual, failing to evaluate candidate resumes against modern quantifiable achievement standards, dynamically adapt daily study schedules to variable time budgets, maintain multi-session contextual memory, or provide proactive autonomous follow-ups to mitigate student procrastination.

Objectives & Expected Outcomes

Core Objectives (5 Points)

- **1. Multi-Session Conversational Mentor:** Build a ChatGPT/Claude-tier chatbot supporting speech-to-text (🎤), file uploads (📎), and STAR behavioral interview coaching.
- **2. Algorithmic Resume ATS Analyzer:** Parse PDF/DOCX to compute an ATS score (0–100) and generate Google X-Y-Z achievement formula rewrites.
- **3. Human + AI Co-Planner:** Construct a calendar scheduler with workload risk diagnosis (Pros & Cons) and 1-click schedule optimization.
- **4. Knowledge Vault & Compatibility:** Build an interactive skill-gap matrix calculating dream company compatibility (Google, TCS) with bridging paths.
- **5. Autonomous Reminder Daemon:** Deploy a background agent sending personalized, localized check-in emails via Resend / SMTP across 4 languages.

Expected Outcomes (4 Points)

- **1. Elevate Candidate ATS Pass-Rate:** Transition candidate resumes from sub-50% rejection baselines to >85% through quantifiable impact metrics.
- **2. Sustain Study Velocity via Focus Budgeting:** Increase curriculum completion by replacing rigid hours with 57-minute or 30-minute focus intervals.
- **3. Democratize Mentorship via Indic Localization:** Eliminate language barriers by providing full technical guidance in English, Hindi, Marathi, and Sanskrit.
- **4. Zero-Friction Access & High Cloud Resilience:** Enable 1-click Demo account login, direct email access, and self-healing session recovery across cloud restarts.

Scope of the Project

In-Scope Deliverables (Pointwise)

- Full-stack responsive web platform built with React 19, Tailwind CSS, and Node.js / Express.
- Dual-engine database architecture: PostgreSQL on Render with SQLite3 local fallback.
- Multi-session chat persistence (chat_sessions, chat_messages) with CRUD operations.
- In-browser Speech-to-Text via Web Speech API in English, Hindi, and Marathi.
- Resume AST parsing via pdf-parse and mammoth for PDF and DOCX formats.
- Autonomous 2-hour study check-in alerts and transactional email dispatches.
- Interactive SVG Donut charts with cursor hover tooltips and dynamic time budgets.

Out-of-Scope (Future Work)

- Real-time video webcam facial emotion and posture analysis (scheduled for Capstone-II).
- Native compiled iOS and Android binary mobile apps (platform is responsive PWA).
- Automated web-scraping or bot applications to external job boards (restricted by TOS).
- Live peer-to-peer mock video calling (system focuses on autonomous AI-to-student coaching).

Review of Existing Literature

Sr.	Paper Title & Ref.	Author Names	Key Conclusion / Methodology	Research Gaps Identified
[1]	Automated Resume Screening Using NLP (IEEE 2022)	S. Roy, A. Narahari et al.	Extracted skill tokens using TF-IDF and Cosine Similarity to match job profiles.	Ignores Google X-Y-Z action metrics; lacks actionable line-by-line rewrite suggestions.
[2]	Conversational AI Mentors in Higher Ed (ACM 2023)	M. Griffiths, L. Chen	Evaluated LLM-driven tutors; proved significant engagement via contextual dialogue.	No persistent multi-session memory; monolingual English only; no file attachment support.
[3]	Personalized Learning Path Generation (Springer 2021)	H. Zhang, T. Wang et al.	Generated prerequisite-aware roadmaps using knowledge graph traversals.	Rigid time blocks; cannot adapt to variable 30m vs 57m daily student availability.
[4]	ATS Formatting Pitfalls & Parser Bias (IEEE 2023)	J. Martinez, P. Sengupta	Demonstrated 68% failure rate on two-column and graphical resume layouts.	Focused solely on parsing diagnostics; did not provide automated skill bridging.
[5]	Human-AI Co-Creation in Time Mgmt (IEEE 2022)	E. Horvitz, R. Patel	Mixed-initiative task planning yielded higher user compliance than automated schedulers.	Lacked automated "Pros & Cons / Workload Risk" diagnostic analyzer.
[6]	Multilingual Prompting in Indic LLMs (IEEE Access 2024)	A. Joshi, R. Kulkarni et al.	Showed necessity of code-switching for technical terms in Hindi and Marathi.	Did not apply Indic localization to technical career mentoring or Sanskrit domain vocabularies.
[7]	Proactive Timing in Academic Procrastination (C&E 2021)	D. Larson, C. Meyer	2-hour interval conversational check-ins reduced task procrastination by 41%.	Used intrusive SMS; no integration with an LLM that summarizes remaining tasks.
[8]	Skill Gap Diagnostics for Tech Undergrads (Proc. CS 2023)	K. Verma, S. Deshmukh	Compared academic university syllabi with active industry job postings.	Lacked interactive company compatibility gauges and 1-click bridging roadmaps.

Literature Survey 1: Resume AST & Skill Matching

Base Paper: "Automated Resume Screening and Skill Matching Using Natural Language Processing" (IEEE 2022)

Authors: S. Roy, A. Narahari, and V. K. Singh | **Venue:** IEEE ICCCNT 2022

- **Methodology:** Applied Named Entity Recognition (NER) and sentence-transformers to parse unstructured resume documents and score candidate relevance.
- **Key Findings:** Token stream parsing retains grammatical context far better than standard regex chunking, boosting skill extraction accuracy to 88.4%.
- **Identified Shortcomings:**
 - ❑ The system generated only a passive match percentage without constructive rewriting feedback.
 - ❑ Failed to evaluate quantifiable impact or action verbs required by elite recruiters.
- **Enhancement in CareerMentor AI:**
 - ❑ Evaluates resumes against the **Google X-Y-Z achievement formula**: *"Accomplished [X] as measured by [Y], by doing [Z]"*.
 - ❑ Provides deep section extraction (Skills, Projects, Education) and actionable rewrite suggestions with 1-click PDF export.

Literature Survey 2: Conversational Career Mentors

Paper: "Conversational AI Mentors in Higher Education: A Systematic Review" (ACM 2023)

Authors: M. D. Griffiths and L. Chen | **Venue:** ACM Transactions on Interactive Intelligent Systems

- **Methodology:** Evaluated LLM-driven dialog systems designed to guide university students through technical skill acquisition and interview preparation.
- **Key Findings:** Dialogue-based agents significantly reduce candidate anxiety during mock interviews when structured around proven behavioral frameworks (such as the STAR method).
- **Identified Shortcomings:**
 - ❑ Surveyed platforms maintained only single-session memory; refreshing the browser cleared all conversational context.
 - ❑ Lacked multimodal speech inputs, preventing realistic verbal interview practice.
- **Enhancement in CareerMentor AI:**
 - ❑ Implements **persistent database-backed multi-session chat** (chat_sessions and chat_messages).
 - ❑ Integrates **Web Speech API microphone voice input** (🎤) and document roadmap attachments (📎).

Literature Survey 3: Dynamic Adaptive Scheduling

Paper: "Personalized Learning Path Optimization and Time-Boxed Workload Regulation" (Springer 2021)

Authors: H. Zhang, T. Wang, and K. Liu | **Venue:** Springer Education & Information Technologies

- **Methodology:** Modeled course curriculums as directed graphs, scheduling study milestones according to historical student velocity.
- **Key Findings:** Rigid fixed-hour schedules resulted in catastrophic dropout rates during midterm exam periods; dynamic task recalibration increased course completion by **54%**.
- **Identified Shortcomings:**
 - ❑ The system operated as an opaque black-box; users could not inspect schedule strengths or potential workload overloads.
- **Enhancement in CareerMentor AI:**
 - ❑ Implements science-backed focus session blocks: **30 min, 45 min, 57 min (optimal focus block), and 90 min.**
 - ❑ Introduces a **Plan Analyzer** displaying Strengths (Pros) and Risks (Cons) with 1-click automated AI schedule re-balancing.

Proposed System Architecture

CareerMentor AI bridges literature gaps through a 4-tier decoupled modular architecture:

1. PRESENTATION LAYER (React 19 + Vite + Tailwind CSS - Dark SaaS #0B1220)

Multi-Session Chatbot (🗣️ Voice Mic + 📎 Attach) • Resume ATS Analyzer (0-100 Score + X-Y-Z Tips) • Co-Planner Grid • Knowledge Vault • 4-Lang i18n

↓ HTTP REST / Bearer JWT ↑

2. APPLICATION & CONTROLLER LAYER (Node.js + Express.js REST API)

Self-Healing Auth Middleware • Chat Controller • Resume Controller • Planner Controller • Roadmap Controller • Reminder Agent

↓ External Calls & Queries ↑

Document Parsing

pdf-parse & mammoth (PDF / DOCX AST)

Generative AI Core

Google Gemini 1.5 Pro / Flash SDK

Persistence Tier

PostgreSQL (Cloud) / SQLite3 (Local)

Notification Agent

Resend API & Nodemailer (SMTP)

Fig. 1: Block diagram of the proposed CareerMentor AI system

References

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- [10] OpenAI, "GPT-4 Technical Report: Educational Benchmarks and Language Processing," *arXiv:2303.08774*, Mar. 2023.

Thank You !!!

CareerMentor AI

Empowering engineering undergraduates through personalized, multilingual, 24/7 autonomous career velocity mentorship.

GitHub: github.com/Harshjha76/CareerMentor_AI

Live Demo: careermentor-ai-c36w.onrender.com

Q & A

Questions & Technical Discussion

1. Cloud Session Recovery

JWT claims decoded to automatically self-heal and re-provision user state in DB if cloud container restarts.

2. Hallucination Guardrails

Strict prompt contracts; YouTube masterclasses linked exactly once per weekly milestone on Day 1.

3. Multilingual Fidelity

Indic prompt engineering preserves technical keywords (DSA, SQL) in English while conversing in Hindi/Marathi.

4. 5-Pillar ATS Formula

30% X-Y-Z metrics, 25% technical skills, 20% structure, 15% metrics, 10% readability.